

M002 Extension — Worksheet (Beginner)

Topic: Redstone Trail Solver (3D Cube Maze) · **Course:** Maze Madness · **Level:** Extension (Beginner) ·

Time: about 35 minutes

This is a **challenge** for students who finished the weekly mazes. The agent follows a **redstone trail** through a maze that goes **up and down**, not just flat. You start it with a chat word: type **run**.

1 · Read the Chat Commands

A chat command binds a **word** to an action. When you type that word in chat, the agent does it.

```
when you type "l":
    agent turns left
```

You type **l**. What does the agent do? Circle one: turns left · turns right · jumps


```
when you type "r1":
    agent teleports back to you
```

You type **r1**. Where does the agent go? Circle one: to the goal · back to you · nowhere

2 · Follow the Trail 🔍

The agent reads redstone around it and follows the trail. Read each line and answer.

```
if redstone is below me:  
    move up
```

There is redstone under the agent. What does it do? Circle one: move up · move down · stop


```
if redstone is above me:  
    move down
```

This maze is 3D. Which way does the agent go here? Circle one: up · down

The agent follows the redstone like a path. In one short sentence, what is the redstone for?

3 · Find the Difference 🐛

Each pair shows clean steps first, then a broken version. Circle what's different and write one short sentence about the bug.

Pair A — The agent should **keep going** until the maze is done.

```
# clean
keep doing forever:
  move forward
  if redstone is below me:
    move up
```

```
# buggy
move forward
if redstone is below me:
  move up
```

What is wrong? How many times does the buggy agent check the trail?

4 · Fill the Gap

The agent should go **up** when there is redstone below it. One word is missing. Fill it in from the word bank.

```
if redstone is below me:  
    move _____
```

Word bank: up · down · left

Write the missing word:

5 · Tell Me What You Built 📷

Open the extension world. Type `run` and watch the agent follow the redstone trail to the goal. When it finishes, come back here.

Send a photo or video of the agent at the goal, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

The maze was 3D, so the agent went ...

The redstone trail told the agent to ...

The hardest part was ...

6 · Record Your Walkthrough 🎥

Take a video on your phone (or a parent's phone) while the agent solves the maze. Talk like you are teaching a friend. Try to use these words: **redstone, trail, up, down, 3D**.

1. Show the maze and the redstone trail.
2. Type **run** and show the agent following it.
3. Say one place where the agent went up or down.

Write what you will say in your video. You can read from it while filming.

Submit ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.